INCH-POUND

MIL-C-70574B (AR)
5 March 1992
SUPERSEDING
(See 6.6)

MILITARY SPECIFICATION

CIRCUIT BOARD ASSEMBLY, BUILT IN TEST 2A8 UNIT 2: 9361010

This specification is approved for use by the U.S. Army Armament, Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements and tests for the AN/VPS-2A Radar receiver-transmitter Built-In-Test (BIT) circuit card assembly, herein referred to as the Unit 2 BIT assembly. (See 6.1)

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation. (See 6.2)

SPECIFICATIONS

MILITARY

MIL-F-13926	- Fire Control Materiel, Manufacture and Inspection, General Specification for
MIL-I-45607	- Inspection Equipment, Acquisition, Maintenance and Disposition of

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, U.S. Army ARDEC, ATTN: SMCAR-BAC-S, Picatinny Arsenal, New Jersey 07806-5000 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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STANDARDS

MILITARY

MIL-STD-109 - Quality Assurance Terms and Definitions
MIL-STD-810 - Environmental Test Methods and Engineering Guidelines

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWINGS (see 6.4)

U.S. ARMY ARMAMENT RESEARCH, DEVELOPMENT AND ENGINEERING CENTER (ARDEC)

9361010 - Receiver Transmitter BIT Assembly
9361341 - Test Program Set
9397373 - U2/U4, Interface Device TPS Adapter Board C, Printed Wiring Assembly
9397502 - Cable IDC W2
9397504 - Test Panel IDC W3
12950877 - Requirements for Soldering Electrical and Electronic Assemblies

SPECIAL PACKAGING INSTRUCTION

PS 9361010 - Packaging of Receiver Transmitter Bit Assembly

Copies of drawings, publications and other Government documents required by the contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Non Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of the documents not listed in the DODISS are the issue of the documents cited in the solicitation (see 6.2).

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INSTITUTE FOR INTERCONNECTING AND PACKAGING ELECTRONIC CIRCUITS

ANSI/IPC-DW-425 - Design and End Product Requirements for Discrete Wiring Boards

ELECTRONIC INDUSTRIES ASSOCIATION

EIA RS-422 - Electrical Characteristics of Balanced Voltage Digital Interface Circuits

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of conflict between the text of this document and the references cited herein the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Materials. Materials shall be in accordance with the applicable drawings and specifications.

3.2 Components and assemblies. The components and assemblies shall comply with all requirements specified on Drawing 9361010, all associated drawings, and with all requirements specified in applicable specifications and standards. Where applicable, the Unit 2 BIT assembly shall meet the requirements of MIL-F-13926 and ANSI /IPC-DW-425 as follows:

3.2.1 MIL-F-13926.

a. Dimensions and tolerances
b. Inorganic protective surface finishings
c. Part identification and marking
d. Workmanship

3.2.2 ANSI/IPC-DW-425.

a. ANSI/IPC-DW-425/11 Type C1 plated through connection

3.3 Environmental service conditions.

3.3.1 Vibration. The Unit 2 BIT assembly shall not suffer damage and shall meet the requirements of 3.4 and 3.5 after exposure to sinusoidal vibrations, in each of three mutually perpendicular axes, as follows:

Frequency Range	Amplitude
5 to 15 hertz	0.06 inches double amplitude

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The frequency of vibration shall be swept logarithmically.
The cycling time shall be 120 minutes per axis.
The sweep rate shall be 3.75 minutes ± 10% per decade.
The 5-55-5 hertz sweep time is 7 minutes 35 seconds ± 10%.

3.3.2 Shock. The Unit 2 BIT assembly shall not suffer damage and shall meet the requirements of 3.4 and 3.5 after exposure to a total of 18 impulse shocks consisting of three shock impulses of 15 g's ± 10%, 11 milliseconds ± 10%, half-sine wave applied in each direction of 3 mutually perpendicular axes.

3.3.3 Altitude.

3.3.3.1 Operating. The Unit 2 BIT assembly shall meet the requirements of 3.4 and 3.5 during and after exposure to barometric pressure encountered at 10,000 feet above sea level (20.6 inches of Hg).

3.3.3.2 Non-operating. The Unit 2 BIT assembly shall meet the requirements of 3.4 and 3.5 after exposure for not less than 10 minutes to barometric pressure encountered at 40,000 feet above sea level (5.5 inches of Hg).

3.3.4 Temperature.

3.3.4.1 Operating. The Unit 2 BIT assembly shall meet the requirements of 3.4 and 3.5 when stabilized thermally at any temperature between -40°F and +155°F.

3.3.4.2 Non-operating (storage). The Unit 2 BIT assembly shall not suffer permanent damage when subjected to the constant temperature extremes of -60°F for a period of 24 hours and +160°F for a period of 4 hours (after temperature stabilization), and shall meet the requirements of 3.4 and 3.5 when returned to room ambient.

3.3.5 Humidity (non-operating). The Unit 2 BIT assembly shall meet the requirements of 3.4 and 3.5 after exposure to an atmosphere with a relative humidity in excess of 95 percent for a period of 10 days.

3.4 Electrical. Each conductor shall be electrically continuous as specified on the schematic diagram of the Unit 2 BIT assembly.

3.5 Performance.

3.5.1 Functional description. The Unit 2 BIT assembly shall monitor control and operation of the receiver-transmitter and shall send this data to the Fire Control Processor when the system is in Radar or BIT mode. The Unit 2 BIT assembly shall monitor radiate command, Modulator Power supply (Mod) and High Voltage (HV) overload indications, beam power supply output voltage, standby and operate

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DC voltages, pulse outputs from the diode switch and diode switch driver assembly, the gated and CW 60 MHz signals, and klystron output power level. The Line Replaceable Units (LRU's) monitored by the Unit 2 BIT assembly shall be Microwave Chassis Assembly (2A1), Klystron Power Supply Assembly (2A2), 60 MHz local oscillator (2A3), Transmit Gate Assembly (2A5), Klystron Beam Power Supply Assembly (2A6), Pulse Shaper (2A7), Unit 2 BIT Assembly (2A8) (See Figure 1).

3.5.2 Power requirements. The Unit 2 BIT assembly power shall be input at connector A2J1-20,21 and return at A2J1-23,24 with requirements as follows:

	MIN	TYP	MAX	UNITS
Input Voltage	16.0	24.0	32.0	Vdc
Input Current	----	124	185	mAdc
Input Impedance	23.3	52.4	93.2	Ohms
Input Power	----	3	5.9	Watts
Line Regulation	----	0.1	0.2	Vdc
Ripple Voltage	----	----	0.6	V Peak-Peak

With the above input, the A1U10 and A2U5 dc to dc converter outputs at the test points shall be as follows:

Test Point	Voltage (Vdc)	Ripple (mVrms)
A1J2-9	+5.00 ± 0.25	450 max.
A2J1-16	-15.00 ± 0.25	40 max.
A2J1-18	+15.00 ± 0.25	40 max.

3.5.3 Clock requirements. The Unit 2 BIT assembly clock (500 kHz) signal applied at J1-C (HI) and J1-D (LO) shall be differential with RS-422 voltage levels per 3.5.6. The clock frequency shall be 500 kHz ± 74 Hz.

3.5.4 TTL voltage levels. The TTL voltage levels for inputs and outputs are as follows:

	MIN	TYP	MAX
Voltage Output Low (VOL)	---	0.2	0.4
Voltage Output High (VOH)	2.4	3.4	---
Voltage Input Low (VIL)	---	---	0.7
Voltage Input High (VIH)	2.0	---	---

3.5.5 CMOS voltage levels. The CMOS voltage levels for inputs and outputs are as follows:

	MIN	TYP	MAX
Voltage Output Low (VOL)	---	---	0.45
Voltage Output High (VOH)	2.4	---	----
Voltage Input Low (VIL)	---	---	0.8
Voltage Input High (VIH)	3.5	---	----

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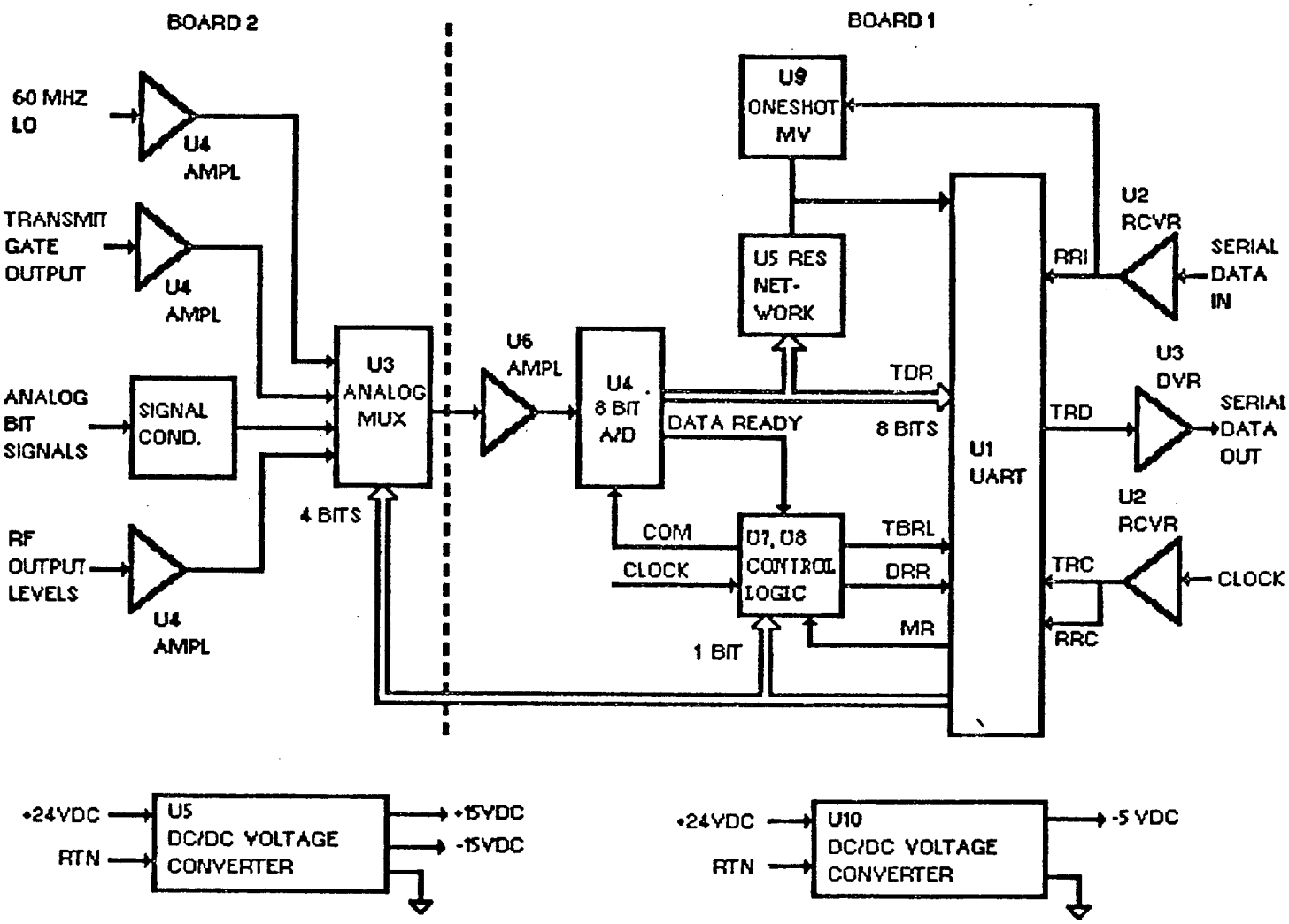


FIGURE 1. Radar Unit 2 BIT Functional Block Diagram.

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3.5.6 RS-422 inputs/outputs.

3.5.6.1 Differential voltages. The RS-422 clock, line receiver and line transmitter buffer differential voltage levels shall be as follows:

	High logic level	Low logic level
Maximum input:	+5.5 V	-5.5 V
Sensitivity:	$V_{ih} = \overline{V_{in}}-V_{in} \geq +0.5$ V	$V_{il} = \overline{V_{in}}-V_{in} \leq -0.5$ V

3.5.6.2 Common mode voltages. The RS-422 common mode voltage levels shall be:

	MIN	TYP	MAX
Voltage Output Low (VOL)	---	---	0.45
Voltage Output High (VOH)	2.5	3.2	----
Voltage Input Low (VIL)	---	---	0.80
Voltage Input High (VIH)	2.0	---	----

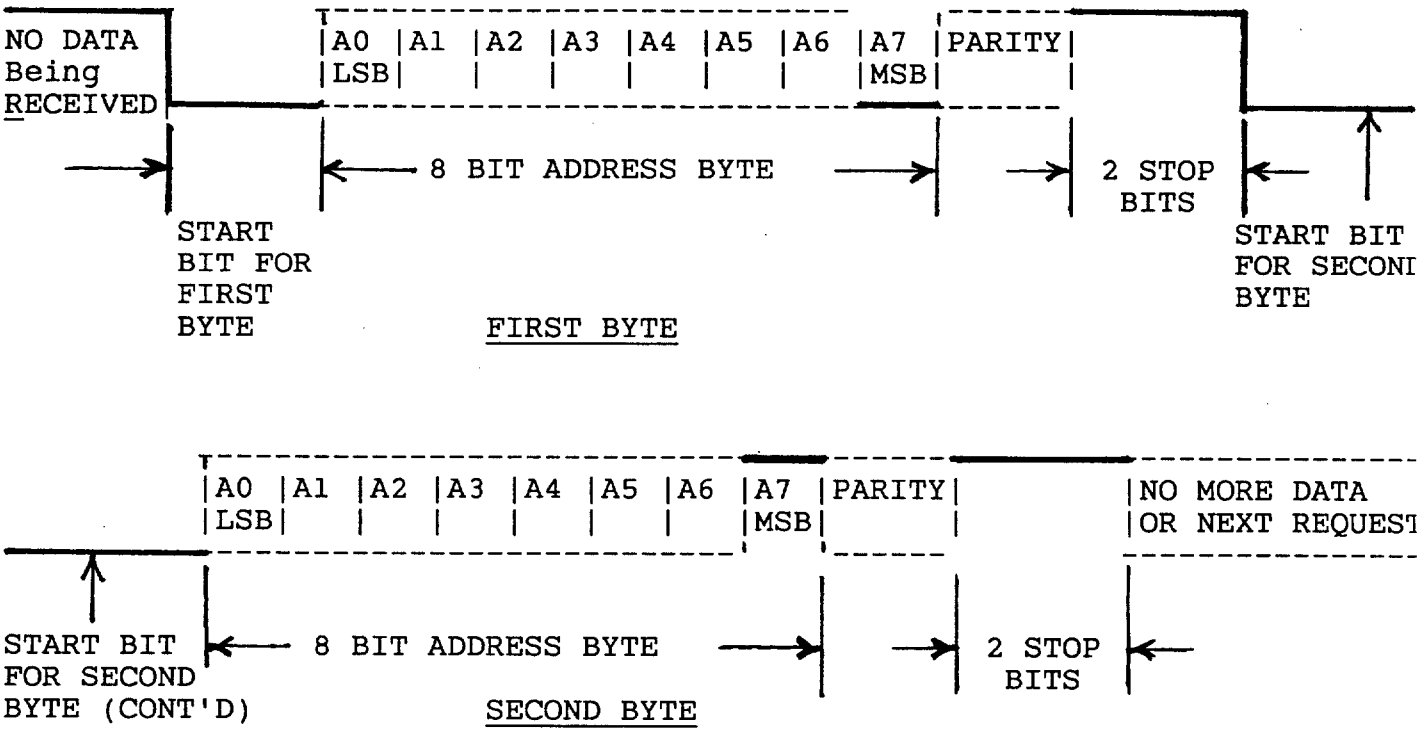
3.5.6.3 Frequencies. The clock and receiver/transmitter data frequencies shall be:

clock - 500.000 kHz ± 74 Hz
data - 31.250 kHz ± 5 Hz

3.5.7 Format of UART A1U1 receiver address bytes. Two address bytes must be received by the A1U1 Unit 2 BIT assembly in order to transmit out the digital byte which represents the requested multiplexer analog input voltage. The first byte has the A7 BIT set low to load the UART's Receiver Buffer Register. The second byte has the A7 BIT set high to cause the low-to-high transition of TBRL, which transfers the data in the UART's Transmitter Buffer Register to the Transmitter Register where it is serially clocked out. The address bytes input into the Unit 2 BIT assembly shall be differential signals with RS-422 voltage levels per 3.5.6 applied at J1-G (HI) and J1-H (LO). With power and clock applied to the Unit 2 BIT assembly per 3.5.2 and 3.5.3 respectively the address data shall have TTL voltage levels per 3.5.4 at test point A1J2-8 (RRI) except that a pull up resistor shall insure the minimum 3.0V CMOS threshold required by the UART. The format for the address bytes shall be per figure 2.

3.5.8 Format of UART A1U1 transmitter outputs. Each byte transmitted from the A1U1 Unit 2 BIT assembly shall be differential with RS-422 voltage levels per 3.5.6, output at J1-J (HI) and J1-K (LO). With power and clock applied to the Unit 2 BIT assembly per 3.5.2 and 3.5.3 respectively, each transmitted output byte shall have CMOS voltage levels per 3.5.5 at A1J2-7 (TRO). The transmitter output byte format shall be per figure 3.

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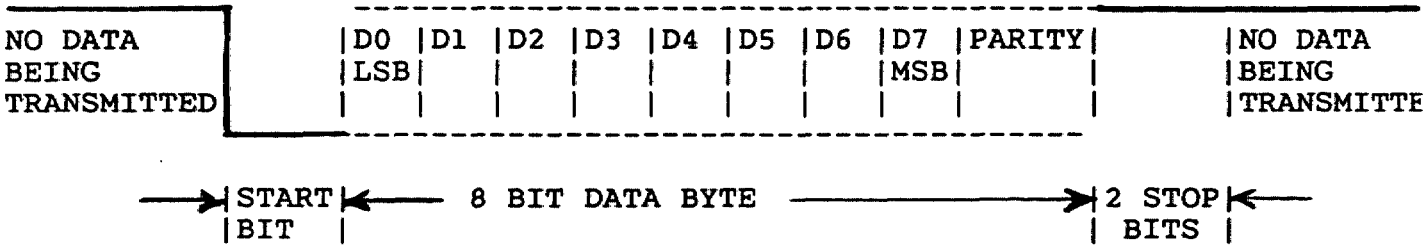


NOTES:

1. The data line is always high when no address bytes are being received.
2. Each start BIT is always low and each stop BIT is always high.
3. The Least Significant BIT (LSB) is the first address BIT received in each byte. The Most Significant BIT (MSB) is the last address BIT received in each byte.
4. When an address word has been received by the UART, it is output on the UART's Receiver Buffer Register (RBR) lines as follows:
   BIT A0 is on RBR1
   BIT A1 is on RBR2
   BIT A2 is on RBR3
   BIT A3 is on RBR4
   BIT A4 is on RBR5
   BIT A5 is on RBR6
   BIT A6 is on RBR7
   BIT A7 is on RBR8
5. Address BITs A4, A5 and A6 are "don't cares" and may be high or low, because RBR lines RBR5, RBR6 and RBR7 are not connected.
6. All BITs of the second address byte are identical to those of the first address byte except for A7 and parity.

FIGURE 2. Format of UART Reciever Address Bytes.

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NOTES:

1. The data line is always high when no data is being transmitted.
2. The start BIT is always low and the 2 stop BITS are always high.
3. The Least Significant BIT (LSB) is the first data BIT transmitted in each byte.
4. The parity BIT is odd; the UART sums data BITs D0 through D7 and adds a one (high parity BIT) or adds a zero (low parity BIT) to cause the sum plus parity BIT to be an odd number.
5. Each data BIT for transmission is input into the UART on the UART's Transmitter Buffer Register (TBR) lines as follows:

BIT 0 is on TBR1
BIT 1 is on TBR2
BIT 2 is on TBR3
BIT 3 is on TBR4
BIT 4 is on TBR5
BIT 5 is on TBR6
BIT 6 is on TBR7
BIT 7 is on TBR8

FIGURE 3. Format of UART Transmitter Output Bytes.

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3.5.9 Master Reset (MR) functions.

3.5.9.1 UART A1U1 not receiving address data. Upon applying power per 3.5.2 and clock input per 3.5.3 to the Unit 2 BIT assembly, when no address data is being clocked into the UART, MR (A1J2-10) shall go high to enable the MR function of the UART. When enabled, the UART's Data Received (DR) line (A1J2-5) shall go low and Transmitter Register Output (TRO) line (A1J2-7) shall go high to initialize the Unit 2 BIT assembly for receiving data and inhibiting data transmission.

3.5.9.2 UART A1U1 receiving address data. With power per 3.5.2 and clock input per 3.5.3 applied to the Unit 2 BIT assembly, when serial address data is being clocked into the UART at CPU TO UNIT 2 DATA with J1-G (HI) and J1-H (LO), MR (A1J2-10) shall go low and disable the MR function of the UART to allow the DR and TRO lines to toggle. The low shall be for a time delay triggered by the high to low transition of the first start bit and extended by every high to low transition (retrigger) within each received byte. Each trigger and retrigger shall begin a 42 $\pm$ 23 millisecond delay (noncumulative), (NOTE: The MR line may self-trigger at cold temperatures due to ripple from the internal 5 Vdc supply. This condition is not considered to be a malfunction) which shall be sufficient to keep the MR line low while address data is being clocked into the UART. After the delay, MR (A1J2-10) and TRO (A1J2-7) shall return high to again initialize the Unit 2 BIT assembly. DR (A1J2-5) shall already be low due to DRR (A1J2-3) going low on the second leading edge of the 500 kHz clock after completion of a data conversion.

3.5.10 Analog multiplexer A2U3, A/D converter A1U4 and UART A1U1 transmitter circuit. Signal timing shall be as shown in figure 4.

3.5.10.1 Initialization. Upon applying power per 3.5.2 and clock input per 3.5.3, the Unit 2 BIT assembly shall initialize (reset) per 3.5.9.1.

3.5.10.2 Multiplexer A2U3 channels. Per Table I,simultaneously applying each Vin to its respective input pin and clocking in the address byte for the Unit 2 BIT assembly A2U3 multiplexer channel S1 per Table I shall cause the following outputs with values per Table I:

a. Received address bits A3, A2, A1, A0 at A2J1-11, 9, 10, 12 respectively.
b. MUX OUT at A2J1-15.
c. A/D converter A1U4 input voltage at pin A1J1-8
d. A/D output byte D7 through D0 at pins A1J1-6, A1J1-4, A1J1-7, A1J1-9, A1J1-3, A1J1-10, A1J1-2, and A1J1-1.

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e. UNIT 2 TO CPU DATA at J1-J (HI) and J1-K (LOW) with format per figure 3.

Similarly, the remaining channels of multiplexer A2U3 shall have all of its outputs as specified in Table I.

3.5.11 Input/Output requirements. The Unit 2 BIT assembly shall have inputs as specified in Table II, and outputs as specified in Table III.

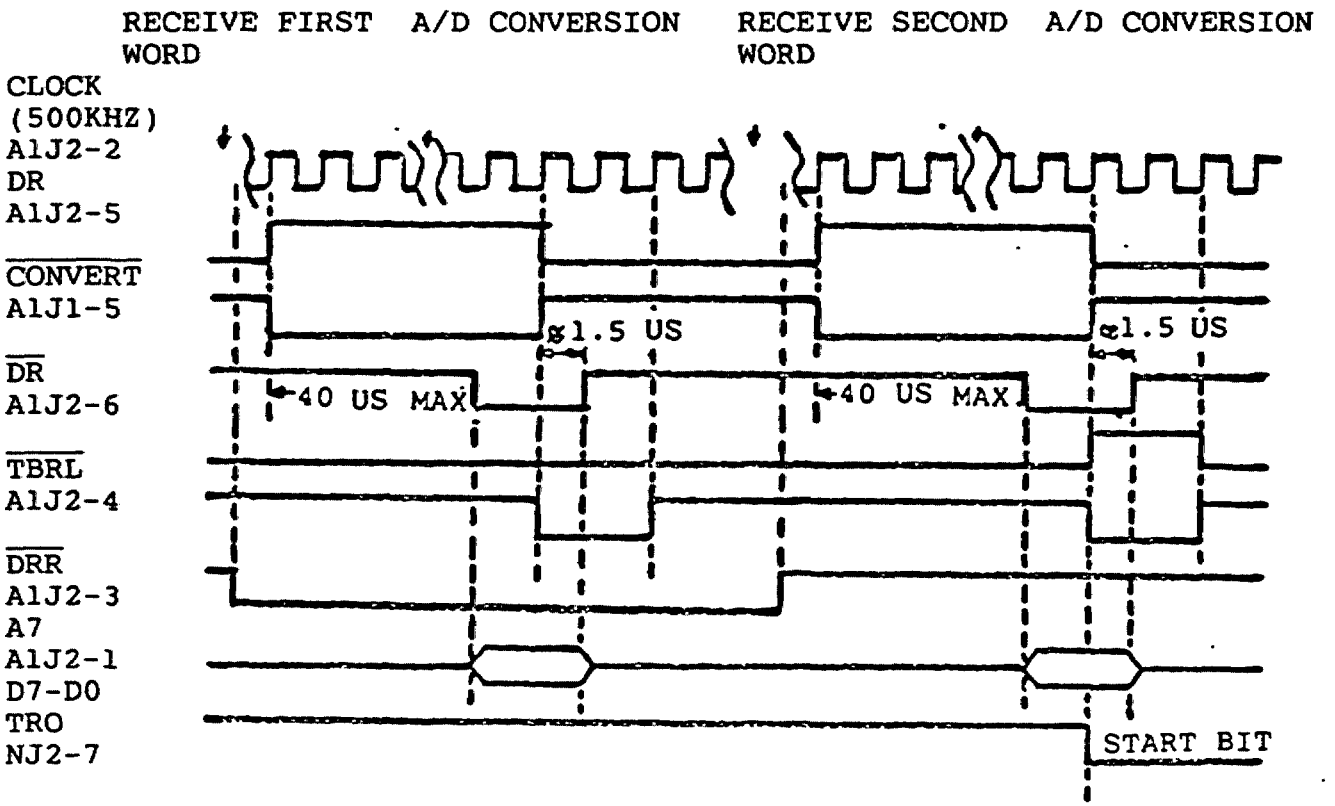


FIGURE 4. Multiplexer, Converter and UART Transmitter Timing Diagram.

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TABLE I. A2U3 Multiplexer Input, A1U4 A/D Converter Input/Output.

A2U3 MULTIPLEX							A2U3 MUX	A1U4	A1U4 A/D Converter Output																
	ADDRESS				CONN.	INPUT		OUTPUT		FROM								TO							Range
CHANNEL	A3	A2	A1	A0	PIN NO.	DESCRIPTION	Vin (Vdc)	(Vdc)	A/D INPUT VOLTAGE (Vdc)	D7	D6	D5	D4	D3	D2	D1	D0	D7	D6	D5	D4	D3	D2	D1	D0
S1	0	0	0	0	J1-7	SWITCH DRIVER OUTPUT	-28.990 TO -29.010	-3.972 TO -4.168	-3.962 TO -4.168	0	0	0	1	1	0	1	1	0	0	0	1	0	0	1	1
S2	0	0	0	1	J1-8	PULSE SHAPER OUTPUT	+19.990 TO +20.010	+2.713 TO +2.875	+2.703 TO +2.875	1	1	0	0	0	1	0	0	1	1	1	0	0	1	0	1	0
S3	0	0	1	0	INTERNAL	-15V	-14.750 TO -15.250	-7.862 TO -8.281	-7.852 TO -8.281	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	1
S4	0	0	1	1	INTERNAL	+15V	+14.750 TO +15.250	+7.862 TO +8.281	+7.852 TO +8.281	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
S5	0	1	0	0	J2	60 KHZ OSC	+ 0.360 TO + 0.400	+3.568 TO +4.109	+3.558 TO +4.109	1	1	0	1	1	0	0	1	1	1	1	0	1	0	1	0
S6	0	1	0	1	INTERNAL	+15V	+14.750 TO +15.250	+1.106 TO +1.190	+1.096 TO +1.190	1	0	0	1	1	0	1	1	1	0	0	1	1	1	1	0
S7	0	1	1	0	J1-13	RF OUTPUT LEVEL	+ 0.036 TO + 0.044	+1.098 TO +1.395	+1.088 TO +1.395	1	0	0	1	1	0	1	1	1	0	1	0	0	1	0	0
S8	0	1	1	1	J3	TRANSMIT GATE OUTPUT	+ 0.004 TO + 0.005	+3.174 TO +4.200	+3.166 TO +4.200	1	1	0	1	0	0	0	0	1	1	1	0	1	1	1	1
S9	1	0	0	0	J1-1	RADAR OPERATE CONTROL	+27.990 TO +28.010	+3.891 TO +4.033	+3.881 TO +4.033	1	1	1	0	0	0	1	0	1	1	1	0	0	1	0	1
S10	1	0	0	1	J1-2	RADIATE COMMAND	+23.990 TO +24.010	+3.335 TO +3.455	+3.325 TO +3.455	1	1	0	1	0	1	0	0	1	1	1	0	1	1	0	0
S11	1	0	1	0	INTERNAL	-15V	-14.750 TO -15.250	-1.106 TO -1.190	-1.096 TO -1.190	0	1	1	0	0	1	0	0	0	1	1	0	0	0	0	1
S12	1	0	1	1	J1-3	+28V STANDBY	+27.990 TO +28.010	+3.891 TO +4.033	+3.881 TO +4.033	1	1	1	0	0	0	1	0	1	1	1	0	1	0	0	0
S13	1	1	0	0	J1-4	MODULAR PWR SUPPLY OVLD	+27.990 TO +28.010	+3.891 TO +4.033	+3.881 TO +4.033	1	1	1	0	0	0	1	0	1	1	1	0	1	0	0	0
S14	1	1	0	1	J1-5	RT PWR SPLY OVLD	+27.990 TO +28.010	+3.891 TO +4.033	+3.881 TO +4.033	1	1	1	0	0	0	1	0	1	1	1	0	1	0	0	0
S15	1	1	1	0	J1-6	BEAM PWR SPLY OUTPUT	-35.990 TO -36.010	-2.698 TO -2.802	-2.688 TO -2.802	0	0	1	1	1	1	0	0	0	0	1	1	0	1	1	1
S16	1	1	1	1	INTERNAL	GROUND	- 0.010 TO + 0.010	-0.010 TO +0.010	-0.020 TO +0.010	0	1	1	1	1	1	1	1	1	0	0	0	0	0	0	0

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TABLE II. Inputs

CONN. PIN #	SIGNAL NAME	SIGNAL
A2J1-1	RADIATE OPERATE CONTROL	+28Vdc ANALOG
A2J1-2	RADIATE COMMAND	+24Vdc ANALOG
A2J1-3	+28V STANDBY	+28Vdc ANALOG
A2J1-4	MODULAR POWER SUPPLY OVLD	+28Vdc ANALOG
A2J1-5	HV POWER SUPPLY OVLD	+28Vdc ANALOG
A2J1-6	BEAM POWER SUPPLY OUTPUT	-36Vdc ANALOG
A2J1-7	SWITCH DRIVER OUTPUT	-29V PULSES
A2J1-8	PULSE SHAPER OUTPUT	+20V PULSES
A2J1-13	RF OUTPUT LEVEL	+.04V
A2J1-14	SHIELD GROUND	----------
A2J1-17	SHIELD GROUND	----------
A2J1-19	SHIELD GROUND	----------
A2J1-20	+24V BATT	PER 3.5.2
A2J1-21	+24V BATT	PER 3.5.2
A2J1-23	+24V BATT RTN	PER 3.5.2
A2J1-24	+24V BATT RTN	PER 3.5.2
A2J2	60 MHz OSC	60 MHz PULSES
A2J3	TRANSMIT GATE OUTPUT	16 MHz PULSES
J1-C	UNIT 2 CLOCK (HI)	PER 3.5.3
J1-D	UNIT 2 CLOCK (LO)	PER 3.5.3
J1-G	CPU TO UNIT 2 DATA (HI)	PER 3.5.6
J1-H	CPU TO UNIT 2 DATA (LO)	PER 3.5.6

TABLE III. Outputs.

PIN #	SIGNAL NAME	SIGNAL
J1-J	UNIT 2 TO CPU DATA (HI)	PER 3.5.6
J1-K	UNIT 2 TO CPU DATA (LO)	PER 3.5.6

3.5.12 Unit 2 BIT Assembly unused pins. The Unit 2 BIT assembly unused pins shall be as listed in Table IV.

TABLE IV. Unused Pins.

CONNECTOR PIN #
A2J1-22
A2J1-25
J1-A
J1-B
J1-E
J1-F

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3.5.13 Test points. The Unit 2 BIT assembly shall have test points as specified in Table V.

TABLE V. Test Points.

CONN. PIN #	SIGNAL NAME	SIGNAL
A1J1-1	D0	PER 3.5.4
A1J1-2	D1	PER 3.5.4
A1J1-3	D3	PER 3.5.4
A1J1-4	D6	PER 3.5.4
A1J1-5	CONVERT	PER 3.5.4
A1J1-6	D7	PER 3.5.4
A1J1-7	D5	PER 3.5.4
A1J1-8	A/D IN	PER 3.5.4
A1J1-9	D4	PER 3.5.4
A1J1-10	D2	PER 3.5.4
A1J2-1	A7	PER 3.5.5
A1J2-2	CLOCK	PER 3.5.3
A1J2-3	DRR	PER 3.5.4
A1J2-4	TBRL	PER 3.5.4
A1J2-5	DR	PER 3.5.4
A1J2-6	DR	PER 3.5.4
A1J2-7	TRO	PER 3.5.4
A1J2-8	RRI	PER 3.5.4
A1J2-9	+5V	PER 3.5.2
A1J2-10	MR	PER 3.5.4
A2J1-9	A2	PER 3.5.5
A2J1-10	A1	PER 3.5.5
A2J1-11	A3	PER 3.5.5
A2J1-12	A0	PER 3.5.5
A2J1-15	MUX OUT	-5V TO +5V
A2J1-16	-15V	PER 3.5.2
A2J1-18	+15V	PER 3.5.2

3.6 First article. When specified in the contract or purchase order (see 6.2), a sample shall be subject to first article inspection in accordance with the technical provisions herein (see 4.3).

3.7 Soldering. All soldering shall be in accordance with the requirements of Note 2 of drawing 12950877.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the

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contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspections set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspections, as part of manufacturing operations,is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 General provisions. The component and subassembly inspection requirements of MIL-F-13926 form a part of the quality assurance provisions of this specification. Reference shall be made to MIL-STD-109 to define quality assurance terms used herein.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).
c. Special sampling inspection (see 4.5).

4.2.1 Inspection conditions. Unless otherwise specified, all examinations and tests shall be conducted at the following standard ambient conditions:

a.	Temperature	+50 deg to +90 deg F.
b.	Relative humidity	90% RH Max.
c.	Pressure	28 to 32 inches of Hg.

4.3 First article inspection.

4.3.1 First article submission. The contractor shall submit a first article sample as designated by the contracting officer consisting of three Unit 2 BIT assemblies for inspection in accordance with 4.3.2.

4.3.2 Inspections to be performed. First article assemblies, components and test specimens may be subjected by the Government to any or all of the examinations and tests specified in Table VI (Functional Test) and Table VII (Special Sampling) herein and to any or all requirements of the applicable drawings.

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4.3.3 Rejection. If any assembly, component or test specimen fails to comply with any of the applicable requirements, the first article sample shall be rejected. The Government reserves the right to terminate its inspection upon any failure of any assembly component or test specimen to comply with any of the stated requirements.

4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. The term "inspection lot" is defined as a homogeneous collection of units of product from which a representative sample is drawn or which is inspected 100 percent to determine conformance with applicable requirements. Units of product selected for inspection shall represent only the inspection lot from which they are drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer, in one unchanged process, using the same materials and methods, in accordance with the same drawings, same drawing revisions, same specifications and same specification revisions, and complies with the provisions for submission of product. All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein, regardless of the type of inspection procedure which is being applied to determine conformance with requirements.

4.4.2 Examination and tests.

4.4.2.1 Components, assemblies and subassemblies.

a. The contractor's quality program or detailed inspection plan shall provide assurance of compliance of all characteristics with the applicable drawings and specifications utilizing as a minimum the conformance criteria specified in this document and the Quality Assurance Provisions (QAP) listed in the Technical Data Package (TDP). In the absence of QAP's, the applicable quality assurance provisions of MIL-F-13926 shall apply.

b. Alternative inspection procedures, methods or equipment, such as statistical process control, tool control, other types of sampling procedures, etc., may be used by the contractor when they provide, as a minimum, the level of quality assurance required by the provisions specified herein. Prior to applying such alternative procedures, methods, or equipment, the contractor shall describe them in a written proposal submitted to the procuring contracting officer for evaluation and approval by the Government. When required, the contractor shall demonstrate that the effectiveness of the proposed alternative(s) is equal to or better than the specified quality assurance provisions herein. In cases of dispute as to whether the contractor's proposed alternative(s) provide equal assurance, the provisions of this specification shall apply. All approved alternative inspection provisions shall be

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specifically incorporated into the contractor's quality program plan or detailed inspection system, as applicable.

4.4.2.2 Final acceptance inspection. Subsequent to first article approval, examinations and tests related to Section 3 shall be performed specified in 4.4.2.3.

4.4.2.3 Functional tests. The tests in Table VI shall be performed on a 100% basis.

TABLE VI. Functional Tests
(100% Inspection)

Characteristic		Requirement	Test Procedure
CRITICAL:		None	N/A
MAJOR:			
101	Electrical	3.4	4.7.2
102	Performance	3.5	4.7.3
MINOR:		None	N/A

4.5 Special sampling. One Unit 2 BIT assembly out of the first 50 produced and one out of each 100 produced thereafter shall be selected at random. The sample shall have met the requirements and tests of Table VI and shall then meet the tests in Table VII. Unless otherwise specified in the contract, the items shall be returned to the lot upon successful completion of testing.

TABLE VII. Special sampling

Characteristic		Requirements	Test Procedure
CRITICAL:		None	N/A
MAJOR:			
103	Vibration	3.3.1	4.7.1.1
104	Shock	3.3.2	4.7.1.2
105	Altitude	3.3.3	4.7.1.3
106	Temperature	3.3.4	4.7.1.4
107	Humidity	3.3.5	4.7.1.5
MINOR:		None	N/A

4.5.1 Failure of sample. Should any one item of a special sampling fail to meet any of the specified test requirements, acceptance of the product shall be suspended by the Government until necessary corrections to all production since the last test as well as to current production has been accomplished in accordance with the provisions of the contract.

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4.6 Inspection equipment.

4.6.1 Government-furnished inspection equipment. Where the contract provides for Government-furnished test equipment, supply and maintenance of test equipment shall be in accordance with the applicable requirements of MIL-I-45607.

4.6.2 Contractor-furnished inspection equipment.

4.6.2.1 Government design. Unless otherwise specified in the contract, all inspection equipment identified by drawing number in specification or QAPs forming a part of the contract shall be supplied by the contractor in accordance with the design specified. The contractor may, however, propose alternatives to Government designs for approval in accordance with 6.3.

4.6.2.2 Contractor design. The contractor shall design and supply inspection equipment compatible with the "Methods of Inspection" specified in 4.7 of this specification and with the component inspection procedures specified in "Examination" and "Test Facilities" requirements of MIL-F-13926 whenever Government designs are not specified (see 4.6.2.1). Since tolerance of test equipment is normally considered to be within 10 percent of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, material, dimensions, and tolerances used in the design of test equipment shall be so selected and controlled as to ensure that the test equipment will reliably indicate acceptability of a product which does not exceed 90 percent of the prescribed tolerance limit and permit positive rejection when nonconforming. Construction shall be such as to facilitate routine calibration of test equipment. Contractor inspection equipment designs shall be submitted for approval in accordance with 6.3.

4.6.2.3 Test equipment. In conjunction with 4.6.2.1 and 4.6.2.2 the test equipment listed in Table VIII shall be used in the performance of the applicable test methods and procedures as specified in 4.7. The following items are required to test the Unit 2 BIT assembly and shall be assembled at the AN/USM-465A test station prior to test execution. Proper operation of interface devices and test equipment shall be verified before using same to run diagnostic tests.

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TABLE VIII. Test equipment list.

	Item	Description
1.	Interface Device	9397373
2.	Cable IDC W2	9397502
3.	Test Panel IDC W3	9397504
4.	Digital Card Tester	AN/USM-465A
5.	ComCode Program Tape	Version 2.0
6.	Test Program Set Manual	9361341
7.	Test Program Set Tape	TPT9361341
8.	Oscilloscope	Input impedance of 1 Mohm or greater, Single sweep capability, with probe, ± 100 Vdc and 500 Hz capability

4.7 Methods of inspection.

4.7.1 Environmental. Environmental tests shall be conducted in accordance with the test conditions, general test performance guidance, and individual test methods of MIL-STD-810, except as specified herein. Test facilities, chambers and apparatus used in conducting the tests contained in the specification shall be capable of meeting the conditions required.

4.7.1.1 Vibration. The test shall be performed as specified in Method 514, Category 8, Procedure I, of MIL-STD-810 as modified by 3.3.1 herein. In Step 1 of Procedure I only steps 4 through 8 of paragraph II-2.1 apply. The Unit 2 BIT assembly shall be attached to the vibration generator directly or with a fixture, and securely held by its normal means of attachment. After vibration, the assembly shall be tested as specified in 4.7.2 and 4.7.3 herein to determine compliance with 3.3.1.

4.7.1.2 Shock. This test shall be performed as specified in Procedure I, Section II of Method 516 of MIL-STD-810 except that:

a. The post test functional check (Step 5 of Procedure I) need not be conducted until completion of all shock impulses.

b. The shock pulses shall be 15 g's ± 10% amplitude and 11 ms ± 10% duration.

c. The maximum g shall occur at 5.5 milliseconds ± 10%.

d. The shock measuring system shall have a flat frequency response with ± 10% from 20 to 2,000 hertz.

After the application of all the shock impulses, the Unit 2 BIT assembly shall be tested as specified in 4.7.2 and 4.7.3 to determine compliance with 3.3.2.

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4.7.1.3 Altitude (operating and non-operating). This test shall be performed as specified in Procedures II and III, Method 500 of MIL-STD-810. The operational checkout shall be as specified in 4.7.2 and 4.7.3 for compliance with 3.3.3.

4.7.1.4 Temperature.

4.7.1.4.1 High (operating and non-operating). This test shall be performed as specified in Procedures I and II, Method 501 of MIL-STD-810 with constant temperature extremes as specified in 3.3.4. The operational checkout shall be as specified in 4.7.2 and 4.7.3 for compliance with 3.3.4.

4.7.1.4.2 Low (operating and non-operating). This test shall be performed as specified in Procedures I and II, Method 502 of MIL-STD-810 with constant temperature extremes as specified in 3.3.4. The operational checkout shall be as specified in 4.7.2 and 4.7.3 for compliance with 3.3.4.

4.7.1.5 Humidity (non-operating). This test shall be performed in accordance with Procedure III, Method 507 of MIL-STD-810 except that in Step 7, checkout shall be performed at the conclusion of the test only, with the item at standard ambient conditions. The operational checkout shall be as specified in 4.7.2 and 4.7.3 for compliance with 3.3.5.

4.7.2 Electrical. Continuity shall be inspected by standard measuring equipment and visual examination for compliance with 3.4.

4.7.3 Performance. The performance of the Unit 2 BIT assembly shall be tested as follows, using the test equipment listed in Table VIII, in accordance with the Unit 2 BIT Test Program Set, drawing 9361341.

4.8 Soldering. Soldering processes shall be inspected by certified inspectors to the requirements and acceptance criteria of note 2 of drawing 12950877.

5. PACKAGING

5.1 General. Preservation, packing, and marking shall be in accordance with Packaging Data Sheet PS9361010. The level of protection shall be as specified in the procurement document.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The Unit 2 BIT assembly shall be used in the RT-860A/VPS-2A Radar Receiver Transmitter in the VADS self-propelled and towed vehicles.

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6.2 Acquisition requirement. Acquisition documents must specify the following:

a. Title, number and date of this specification.
b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1).
c. Applicable stock number.
d. First article sample requirements.
e. Packaging requirements, if other than specified in Section 5.
f. Serialization requirements, if applicable.
g. Certification of conformance for each lot of shipment of product.

6.3 Submission of contractor inspection equipment designs for approval. Submit two copies of designs as required to: Commander, U.S. Army AMCCOM, ATTN: SMCAR-QAF-I (D), Picatinny Arsenal, NJ 07806-5000. This address will be specified on the Contract Data Requirements List, DD Form 1423 in the contract.

6.4 Drawings. Drawing listed in Section 2 of this specification under the heading U.S. Army Armament Research, Development and Engineering Center (ARDEC), may also include drawings prepared by, and identified as, U.S. Army Armament Research and Development Command (ARRADCOM), Frankford Arsenal, Rock Island Arsenal or Picatinny Arsenal drawings. Technical data originally prepared by these activities is now under cognizance of ARDEC.

6.5 Key word listing.

Processor, Fire Control
Receiver Transmitter
Test Program Set (TPS)

6.6 Changes from previous issue. The previous issues were for ARDEC internal use only. They were not forwarded to Naval Publication and Forms Center (NPFC) for publication. This issue is the initial public release of unlimited distribution.

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3. DOCUMENT TITLE
CIRCUIT BOARD ASSEMBLY, BUILT IN TEST 2A8 UNIT 2: 9361010

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets if needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
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